- **HUMAN ASPECTS OF SOFTWARE ENGINEERING**

Slide Set to accompany

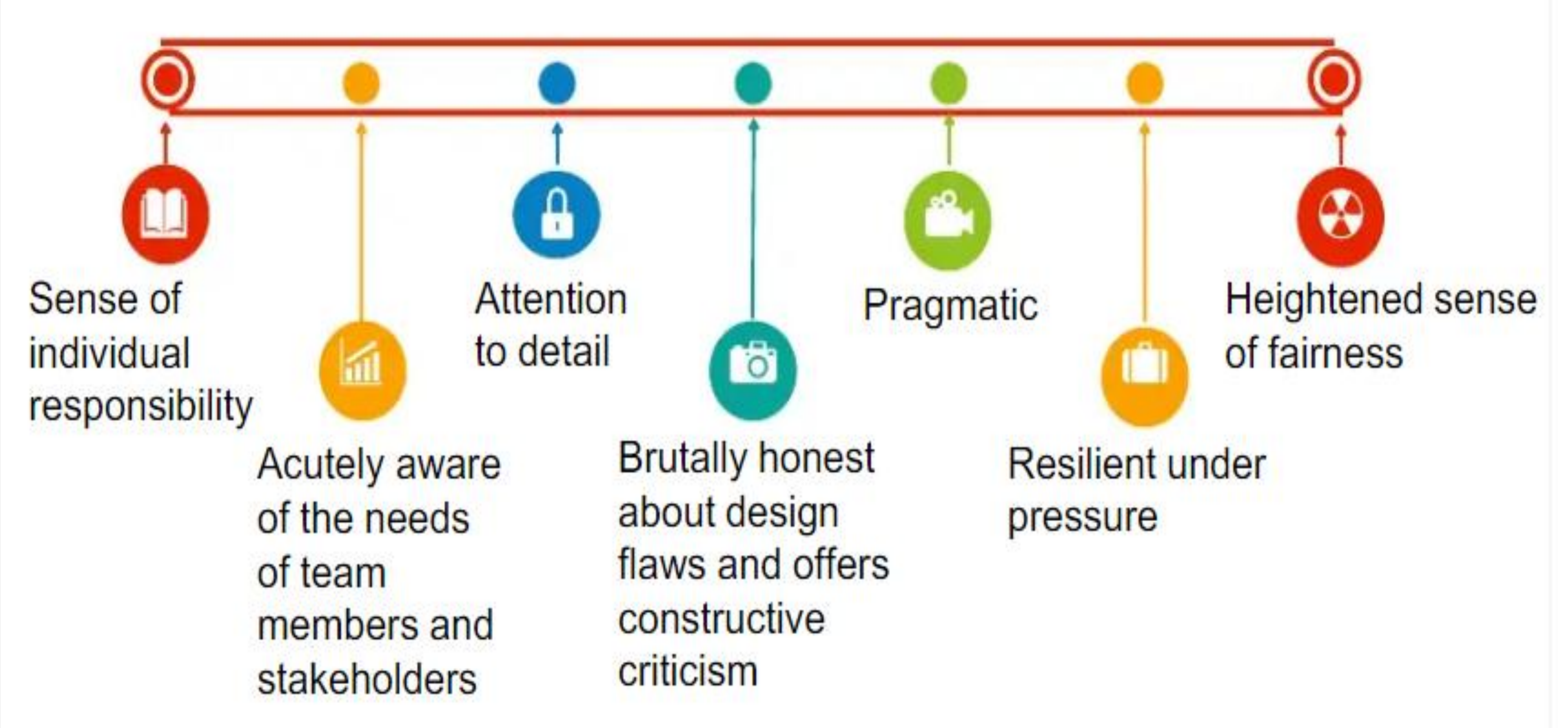
Software Engineering: A Practitioner's Approach, 7/e

by Roger S. Pressman

HUMAN ASPECTS OF SOFTWARE ENGINEERING

- ***Software is developed by people, used by people, and supports interaction among people.***
- ***human characteristics, behavior, and cooperation are central to practical software development.***
- ***Erdogmus** [Erd09] identifies **seven traits:***

Traits of Successful Software Engineers



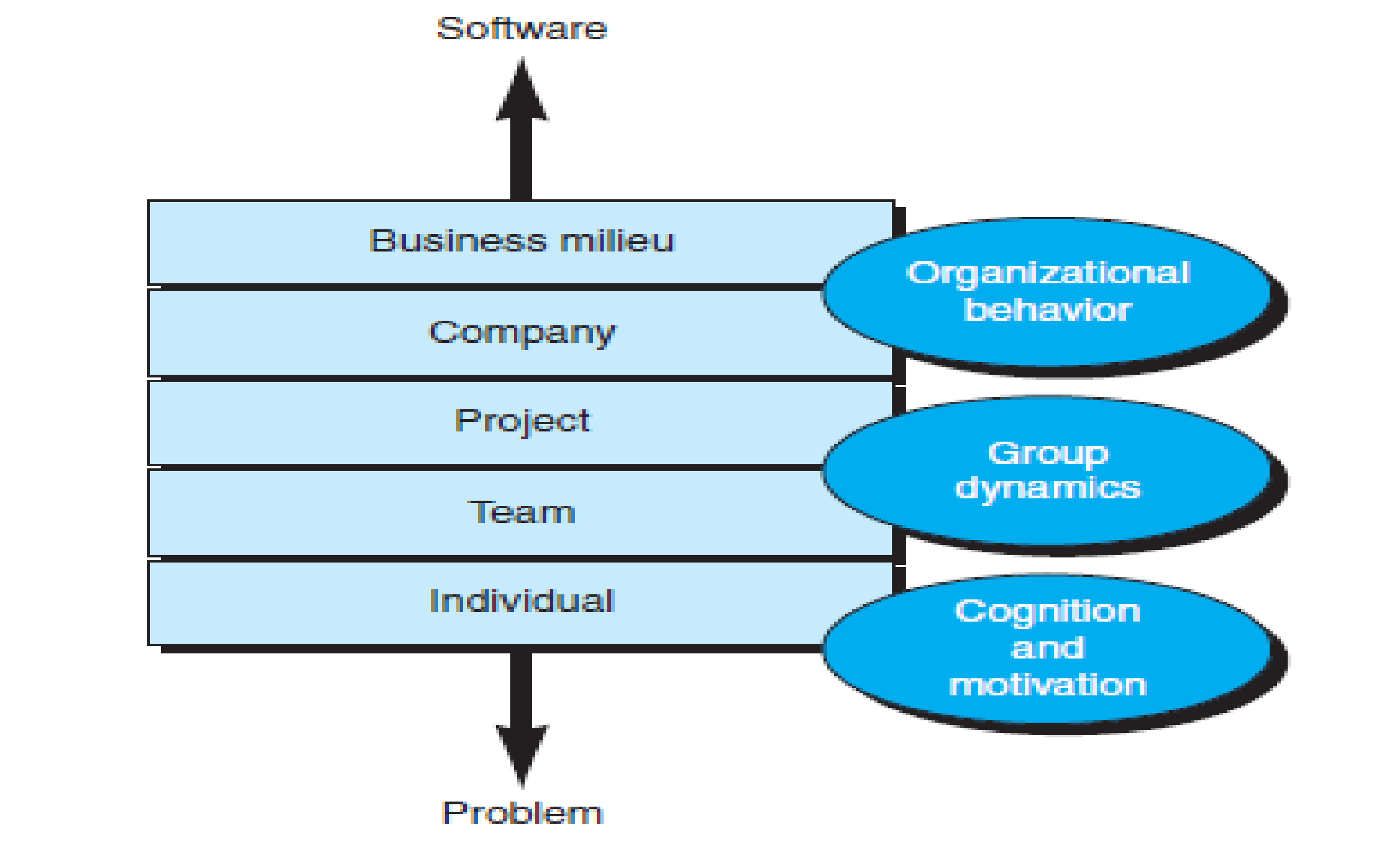
Class Discussion

- How can you be **"brutally honest"** and still not be perceived (by others) as insulting or aggressive?

Class Discussion

- Based on your personal observation of people who are excellent software developers, mention one personality trait that appears to be common among them and one that is uncommon.
- Are there any other personality traits you think should be mentioned?

Behavioral Model for Software Engineering



Behavioral Model for Software Engineering

At the team level, **Sawyer and his colleagues** [Saw08] suggest that:

- Teams often establish **artificial boundaries**
  - reduce communication and, as a consequence, reduce the team's effectiveness.

Boundary Spanning Team Roles

Effective Software Team Attributes

Tom DeMarco and Tim Lister [DeM98] discuss the cohesiveness of a software team:

- **Team** --> What is missing is a phenomenon that we call **jell**.
- **A jelled team** is a group of people so strongly knit that the whole is greater than the sum of the parts . . . .

Effective Software Team Attributes

- Sense of purpose
- Sense of involvement
- Sense of trust
- Sense of improvement
- Diversity of team member skill sets

Effective Software Team Attributes

- … not all teams are effective and
- … not all teams jell.
  - Jackman [Jac98] calls “ **team toxicity**

Avoid Team "Toxicity"

A frenzied work atmosphere in which team members waste energy and lose focus on the objectives of the work to be performed.

High frustration caused by personal, business, or technological factors that cause friction among team members.

Unclear definition of roles resulting in a lack of accountability and resultant finger-pointing.

"Fragmented or poorly coordinated procedures" or a poorly defined or improperly chosen process model that becomes a roadblock to accomplishment.

"Continuous and repeated exposure to failure" that leads to a loss of confidence and a lowering of morale.

Class Discussion

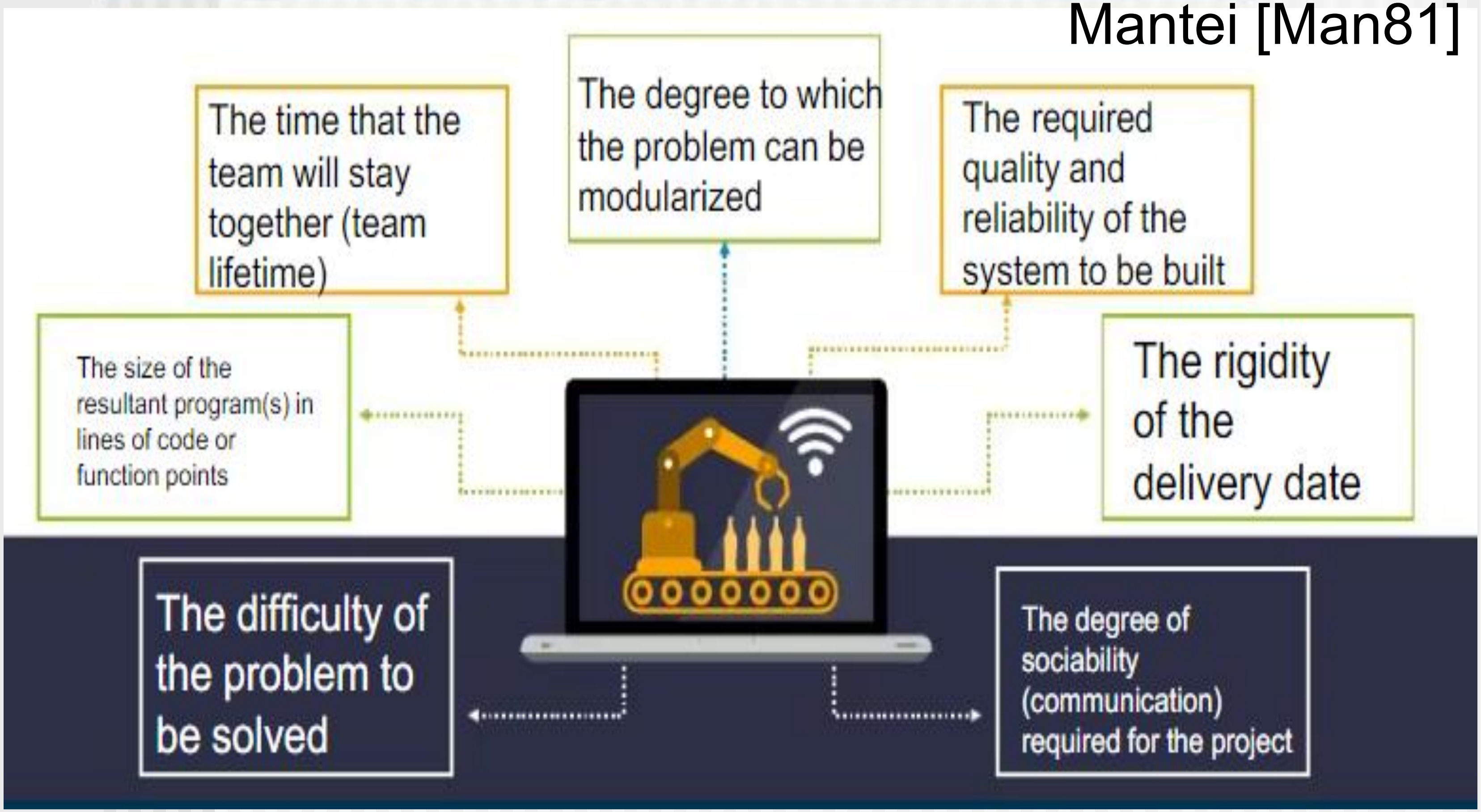
- Based on your personal observation of people during software development or group assignment, identify one team's toxicity.

Class Discussion

- Are extroversion and introversion considered negative traits?
- What insights do you have on human individual differences?

Factors Affecting Team Structure

Mantei [Man81]



Organizational Paradigms

- closed paradigm—structures a team along a traditional hierarchy of authority
- random paradigm—structures a team loosely and depends on individual initiative of the team members
- open paradigm—attempts to structure a team in a manner that achieves some of the controls associated with the closed paradigm but also much of the innovation that occurs when using the random paradigm
- synchronous paradigm—relies on the natural compartmentalization of a problem and organizes team members to work on pieces of the problem with little active communication among themselves

suggested by Constantine [Con93]

Generic Agile Teams

- Stress individual competency coupled with group collaboration as critical success factors
- People trump process and politics can trump people
- Agile teams as self-organizing and have many structures
  - An adaptive team structure
  - Uses elements of Constantine's random, open, and synchronous structures
  - Significant autonomy
- Planning is kept to a minimum and constrained only by business requirements and organizational standards

XP Team Values

- Communication – close informal verbal communication among team members and stakeholders and establishing meaning for metaphors as part of continuous feedback
- Simplicity – design for immediate needs nor future needs
- Feedback – derives from the implemented software, the customer, and other team members
- Courage – the discipline to resist pressure to design for unspecified future requirements
- Respect – among team members and stakeholders

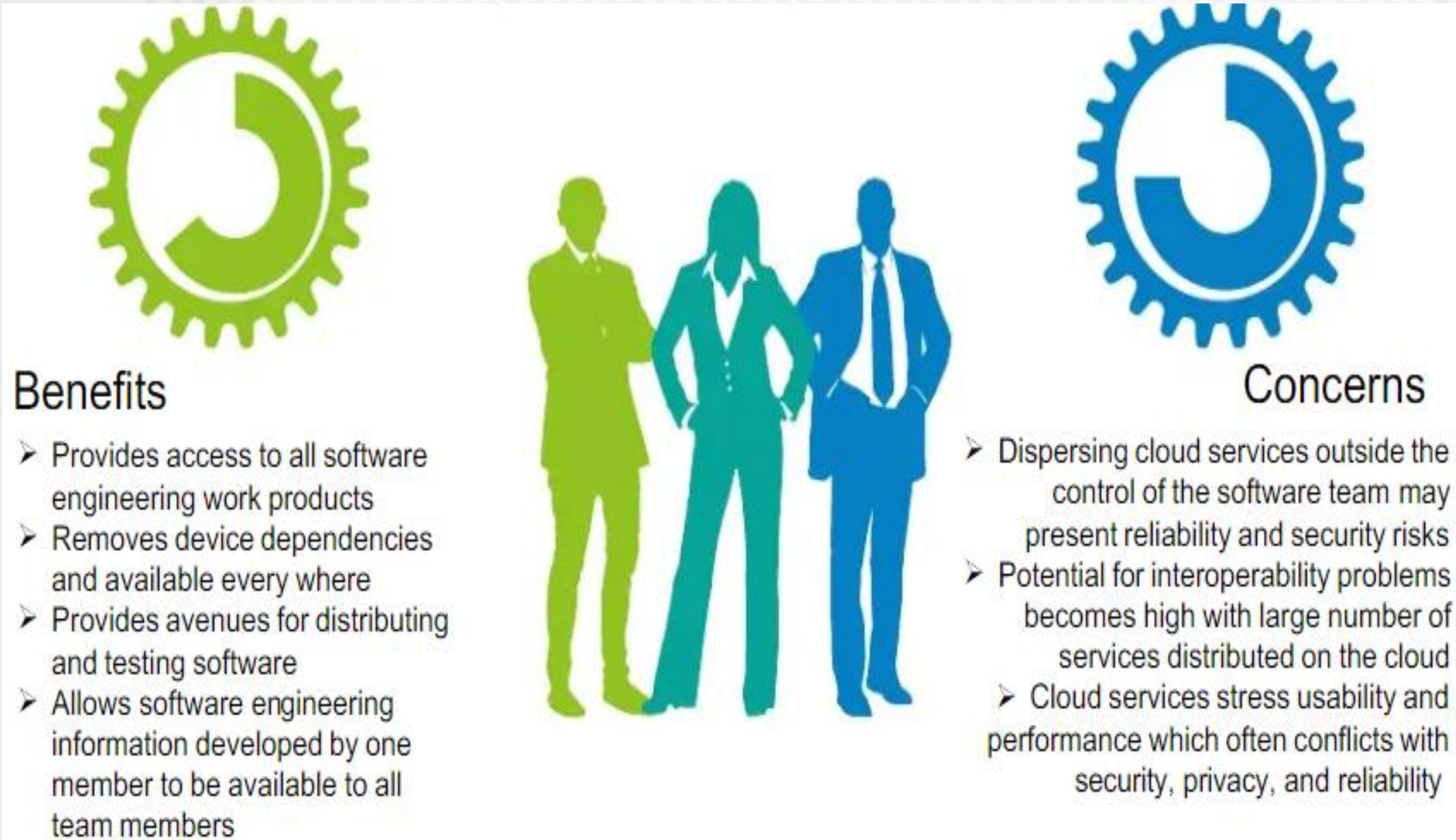
Beck [Bec04a]

Class Discussion

- Privacy and security issues should not be overlooked when using **social media** for software engineering work. Discuss.

- Discuss the benefits and concerns of using the cloud for software engineering work.

Software Engineering using the Cloud



Collaboration Tools

- Software Development Environments (SDEs) of the last century
- *Vs*
- ***Collaborative development environments* (CDEs)**

- How have Software Development Environments evolved from being 'personal productivity tools' of the last century to becoming 'ecosystems of collaboration' in the age of CDEs, and what has been lost—or gained—in that transition?

Collaboration Tools

- Namespace that allows secure, private storage or work products
- Calendar for coordinating project events
- Templates that allow team members to create artifacts that have common look and feel
- Metrics support to allow quantitative assessment of each team member's contributions
- Communication analysis to track messages and isolates patterns that may imply issues to resolve
- Artifact clustering showing work product dependencies

Team Decisions Making Complications

- Problem complexity
- Uncertainty and risk associated with the decision
- Work associated with decision has unintended effect on another project object (law of unintended consequences)
- Different views of the problem lead to different conclusions about the way forward
- Global software teams face additional challenges associated with collaboration, coordination, and coordination difficulties

Factors Affecting Global Software Development Team